

# Traceability in the Supply Chain

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## ABSTRACT

Traceability in firms' supply chain operations has become increasingly important today, calling for greater scrutiny and transparency. Firms have responded by increasing and improving product traceability throughout their global supply. Traceability is a significant benefit to firms. Areas affected include quality control and product safety, tracking product recalls, and reverse logistics. Research does exist on the importance and benefits of implementing traceability initiatives but in very targeted areas. In addition, missing from the literature is the important discussion of what factors predicate firms to implement traceability initiatives beyond those prescribed by law and how industries other than very specific categories, create and implement effective traceability initiatives throughout the supply chain. In turn, the purpose of this research is to investigate traceability to gain greater understanding of why firms implement traceability & what actions or initiatives lead to greater traceability effectiveness.

## KEYWORDS

Contingency Theory, Supply Chain Effectiveness, Supply Chain Management, Traceability

## INTRODUCTION

Global supply chain traceability has become an increasingly important issue in recent years with calls for greater scrutiny and transparency (Steven, 2015; MacCarthy et al., 2016). Government, media, industry groups, suppliers, customers, and firms themselves are all interested in a deeper understanding of how firms manage their global supply chains in terms of information exchange (Silverstro & Lustrato, 2014), quality control (Chen et al., 2014), supplier management (Handfield et al., 2015), disruption avoidance (Kleindorfer & Saad, 2005), and risk management (Grötsch et al., 2013). Recent supply chain safety and security issues in different industries, including restaurant (e.g., Chipotle), pharmaceutical (e.g., Heparin issue 2008), consumer food (e.g., e-coli in hamburger), and energy (e.g., oil rail cars), have further exacerbated demands that firms improve supply chain traceability. As a result, firms have started to increase and improve product traceability throughout their global supply chains in order to have readily available information for internal use and to satisfy stakeholder demands (Shafiq et al., 2014).

A long history of research exists pertaining to agriculture, food networks, and the food supply chain overall (Fearne, 1998; Mattevi & Jones, 2016). Targeted information systems research has

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studied the traceability abilities of software, which tends to focus on version updates (e.g. Ramesh & Jarke, 2001). In the context of these industries, traceability has been extensively studied from different perspectives, including the importance and benefits of implementing traceability initiatives (Fearne, 1998; Mattevi & Jones, 2016). Also, traceability is a significant benefit to firms in terms of managing quality control and product safety, tracking product recalls, and optimizing reverse logistics processes (Chen et al. 2014; Dai et al., 2015). Recently, there has been more research covering tools to effectively and efficiently manage traceability in the supply chain. Information technologies such as RFID have become important enablers for tracing products/services through the supply chain because they promise improved speed, transparency and security of information sharing (Jakkhupan et al., 2015; SAE, 2015; U.S. House of Representatives, 2007). Serialization is another track-and-trace system adopted by the pharmaceutical supply chain to extend the identification of drugs down to the individual level, so that each unit can be traced from its origin (Mallesh et al., 2015).

Yet, for all its purported benefits, traceability can be difficult to implement both internally and across the supply chain. Implementation is difficult because of the complex and dynamic factors that are inherent in today's global business environment. Firms often struggle to obtain critical, accurate, and up to date logistics information from supply chain partners (Mattevi & Jones, 2016). Also, the breadth of the ways companies receive and utilize information differs from firm to firm and supply chain to supply chain (Mattevi & Jones, 2016). Finally, some firms and industries feel significantly greater pressure than others to implement traceability initiatives in their supply chains because [1] they have high product and supply chain complexity and risk; or [2] they are in the sectors with regulations governing documented traceability processes (e.g. pharmaceuticals and food products). Regattieri et al. (2007) propose that a general traceability framework for food products consists of four pillars of product identification, data to trace, product routing and traceability tools. Based upon Regattieri et al.'s framework, Hu et al. (2013) analyze the implementation of a traceability system in the vegetable supply chain in China.

Practices in the food industry often require traceability be considered an integral part of logistics management including reverse flow and storage of goods, services, and related information (Bosona & Gebresenbet, 2013; Kumar et al., 2015). Yet streamlining reverse logistics process has traditionally been seen as a low priority because collection, reuse and reprocessing of returned materials require integration at various levels of a supply chain. When products are moving against the normal flow, process integration becomes a challenge across organizational boundaries. A traceability system can be an effective mechanism for reverse logistics as it [1] facilitates to establish a standard operating procedure so that a company can act quickly to handle a return with reduced costs in reverse logistics (Kumar et al., 2015); and [2] assures visibility and traceability of information and responsive communication among consumers, employees, suppliers, government agencies, and other affected participants along a supply chain. This is critical to efficiency and effectiveness of reverse logistics, and the reputation and image of a company as well (Bosona & Gebresenbet, 2013).

Although previous studies have demonstrated that traceability system implementation is a complex task, the traceability literature still lacks in important discussion of what factors predicate firms to create and implement effective traceability initiatives throughout the supply chain. This is particularly of initiatives beyond those prescribed by law and in specific industries (e.g., food/agriculture/information systems/electronics). Thus, the purpose of this research two-fold: First, to investigate why firms implement traceability, in addition to regulatory mandates, and second, to better understand which actions or initiatives lead to greater traceability effectiveness in a cross-section of firms in diverse industries. We address this dual purpose by using contingency theory to help explain traceability initiatives in firms via five dimensions of traceability: top management support (TMS), nonconforming process management (NCPM), lot traceability (LT), and internal (ITP) and external traceability processes (ETP).

The next section reviews the relevant literature on traceability and introduces the theoretical background of our research, followed by development of the research hypothesis. Then, we discuss

the research methodology, the study, and the results of data analysis. Finally, implications, limitations and possible future research directions are provided.

## **THEORY AND HYPOTHESES DEVELOPMENT**

### **Contingency Theory and Traceability**

Contingency theory considers the impact of environmental factors on an organizational structure, strategic decision making, and efficiency and effectiveness-driven performance criteria (Donaldson, 2001; Ruekert et al., 1985). The theory further suggests that firms will focus internal resources and competencies in order to develop a strategic contingent response to the changing environmental variables in order to remain competitive (Germain et al., 2011; Luthans & Stewart, 1977). The turbulent and uncertain nature of today's global business environment motivates managers to develop a contingent response in the form of strategies which change supply chain structures in order to "fit" with the external environment to remain, or become, competitive (Germain et al., 2011).

The contingency approach explains that managers consider the interrelationships among environmental, management, and performance variables (Luthans & Stewart, 1977). In the context of traceability, firms implement a contingent response to demands on the firm from the external environment (e.g. stakeholder concerns, competitive environment, industry characteristics), the internal environment (e.g. top management directives), with the overall strategy of improving the competitiveness of the firm (Thompson, 1967; Luthans & Stewart, 1977). In line with this thinking, Barratt et al. (2007) explain that implementation of traceability initiatives can enable firms to create unique resources that others may not be able to imitate and differentiate firms from their competitors.

Traceability is defined as "the ability to identify and verify the components and chronology of events surrounding products at all stages of the supply chain" (Skilton & Robinson, 2009, p. 41). Traceability applies to the tracking of products from their raw materials through transformation into a consumer product, and finally distribution. Effective supply chain traceability helps manufacturers feasibly and accurately track raw materials, WIP, and finished goods and services, wherever they may reside in the supply chain. Traceability can be beneficial to consumers, manufacturers, auditors, and other stakeholders involved in the supply chain process through the measurement of product and service history. This allows for supply chain transparency, improves record accuracy, and gives firms the means to maintain close control over their products. The implementation of traceability measures can also improve firms' competitiveness.

In the context of traceability, contingency theory posits that managers are more likely to implement initiatives in industries and competitive environments where product and supply chain complexity and risk are higher (Grötsch et al., 2013). This is because complexity and risk create unique situations in which managers must be proactive and manipulate their resources and strategies to fit with this challenging external environment in order to remain or become competitive (Grötsch et al., 2013).

### **Dimensions of Traceability Effectiveness**

This research adapts elements from four key studies on traceability information models and then distills those elements into five key dimensions of traceability effectiveness. The four studies include Bechini et al. (2008), who introduce a data model for traceability, ECR (2004) research that develops a set of best practices for product traceability, research by Jansen-Vullers et al. (2003) that identifies a framework of four elements of traceability based on various traceability definitions, and Kelepouris et al. (2007), who create a complete traceability model. The five dimensions of traceability effectiveness used in this research and their relation to each of the aforementioned studies are detailed next.

Jansen-Vullers et al. (2003) state traceability activities should, at a minimum, include lot traceability, management involvement, and internal and external reporting for traceability data retrieval. Similarly, ECR (2004) posited four key traceability principles in the same theme: traceability capture

and recording procedures, physical lot traceability, internal management/retrieval of traceability data, and external data communication of traceability data. Requirements other than just legal considerations that drive traceability within the supply chain have also been discussed. These include management buy-in as a strategic goal, specific processes for product traceability to maintain safety and quality, unique lot traceability, and systems to store and allow retrieval of data from both internal and external sources (Bechini et al. 2008). Finally, Kelepouris et al. (2007) furthered the work by Jansen-Vullers et al. (2003) and ECR (2004), comparing the two models and combining the elements into a complete traceability model with five elements.

Adapting concepts the previous studies, we propose five key dimensions that are critical to traceability effectiveness: top management support, nonconforming processes/procedures, lot traceability, internal systems management for traceability, and external systems management for traceability. Individual items for each dimension scale were adapted from Speier et al., (2011). To test the items and scales for face validity, short interviews were conducted with supply chain managers from different industries. These interviews were transcribed and analyzed. The participants' responses were consistent with our five identified dimensions and consistent with the literature in the field of product/service traceability.

### ***Top Management Support***

Full traceability in the supply chain requires participation and support by internal and external supply chain partners, including top management (Kelepouris, 2007). Strategic guidance from top management is needed to commit financial resources to projects, to provide moral support, and to ensure task delegation (Gunasekaran & Ngai, 2004). Furthermore, the relationship between strategy and traceability is a top-down approach; successful traceability initiatives are often initiated and championed by top management. Contingency theory explains that top management is in a position to recognize the demands coming from stakeholders for traceability initiatives, and move them forward as unsupported initiatives can often be ignored or set aside.

### ***Non-Conforming Processes/Procedures and Lot Traceability***

Traceability system must also include formal non-conforming processes/procedures and control mechanisms for lot traceability for success (Bechini, 2007; Merminod & Pache, 2011). Non-Conforming process/procedures and lot traceability represent two similar, but unique constructs. The former focuses on the degree to which a firm has established processes/procedures in place for sharing information in a timely fashion in the case of a traceability issue. The latter, lot traceability, is the specific manifestation of processes/procedures as the mechanism to identify where a quality or abnormal issue originated. Lot traceability allows a firm to understand the origin, routing, and transfer of those products/services in the case of a missed shipment, contamination, etc. (Speier et al., 2011). Together, non-conforming processes/procedures and lot traceability provide a first line of defense by which to prevent, detect, respond, and recover from a product/service traceability issue.

Previous research indicates that elements such as processes/procedures for non-conformance and concrete lot traceability enhance the flow of information between supply chain partners (Blackhurst et al., 2005). In addition, supply chain dynamics can be captured and uncertainty reduced when using these elements for internal and external communication (Merminod & Pache, 2011).

### ***Internal and External Systems Management***

Firms employ internal and external systems and relationships to effectively manage information when full traceability is required (Viaene & Verbeke, 1998). Sarpong (2014) also discusses the issues and concerns behind internal as well as external systems for traceability. Furthermore, there must be interaction between internal and external systems management for supply chain traceability, including processes for how buyers and suppliers transfer responsibility (Kelepouris et al., 2007; Merminod & Pasche, 2011).

### *Internal*

Internal systems management for traceability assess the degree to which a firm has integrated methods for product/service traceability within their organization. Firms with integrated methods for product/service traceability are should be able to identify potential problems early and to better respond should an issue occur (Senneset et al., 2007).

Managers tend to develop and adopt internal traceability systems mainly to improve quality; traceability can be seen as a subsystem whose presence is essential to the management of product/service quality (Peri, 2002). Firms typically establish internal systems for traceability through entities such as ISO or through techniques such as failure modes effects and criticality analysis (FMECA) (Bertolini et al., 2006). For example, the ISO 22005 Traceability Standard requires that each company know their immediate supplier and to whom the product is being sent. ISO 22005 also states that food safety is the joint responsibility of all the providers involved (International Organization for Standardization, 2007). FMECA is a technique developed by Ford Motor Company and the United States Military (Mode, 1988; US Military Standard, 1980; US Military Standard, 1983) that identifies and assesses how potential failures can affect the performance of a process or product.

### *External*

External traceability is described as the processes of physically handing over a traceable product from one traceability partner to another (Veronneau & Roy, 2009). Collaboration with external partners has long been cited as critical factor in any program to ensure visibility and traceability of products and services within a supply chain (Rinehart, et al., 2004; Voss et al., 2009). External systems for traceability include procedures to monitor, collaborate, and define consequences from areas such as contamination, quality, and/or non-conformance. Research by Ellis et al. (2010) and Speier et al. (2011) state that as the level of supply complexity and risk increases, the more likely firms are to select suppliers who are better able to trace products/services throughout the supply chain.

When actors in supply chain reach agreement to share information with each other, the next step is to utilize information sharing to improve supply chain performance. By revealing necessary information to all actors, the actors will clearly find out the sequence of operations and activities and therefore be able to coordinate each other activities along the supply chain (Kärkkäinen et al., 2003). In addition, this enables down-stream firms to monitor and control upper-stream production quality (Lyles et al., 2008).

Lyles et al. (2008) posit that using traceability to monitor and/or control upstream suppliers is mainly to control product/service quality from the upstream perspective. However, the implementation of traceability is not the absolute way of solving operational problems. For example, product quality problems can emerge due to buyer inability to create transparency with their suppliers and detect the origins of mistakes (Lee & Özer, 2007). Also, even with traceability initiatives in place all along the supply chain, sources of failure may still emerge if traceability does not cover all crucial points where problems may occur (Bertolini et al., 2006).

## **Hypothesis Development**

Each of the five dimensions of traceability effectiveness were operationalized and measured using items adopted from four studies: Benchini et al. (2008), ECR (2004), Jansen-Vullers et al. (2003), and Kelepouris et al. (2007). The hypotheses attempt to synthesize key factors across these five dimensions in order to contextualize the dimensions into contingent responses as initiatives that address each dimension issue. From a contingency perspective, the response by managers is an attempt to “fit” strategies to external factors by dedicating resources to specific initiatives and operations (Grötsch et al., 2013). The factors are drawn from Clarke (1997), Lyles et al. (2008), Speier et al. (2011), and Perrow et al. (1984) and were adapted to aid in the development of four key factors that impact a firms’ product/service traceability in the supply chain. The first two are firm ISO certification and supplier ISO certification and represent the internal environment, as ISO certification processes

are ultimately initiated and controlled by the buying firm. Speier et al. (2011) speak to establishing standardized policies via a common certifying body.

The next two factors are supply chain complexity and product/service risk and represent the external environment. Firms that seek to reduce supply chain disruptions do so by actively managing their supply chain complexity (Perrow et al., 1984; Speier et al. 2011). Furthermore, Clarke (1997), Lyles et al. (2008), and Speier et al. (2011) all speak to how firms with higher risk products/services act more cautiously than firms with lower risk products/services. Each of the four factors are described in the following sections, and then related to the five constructs representing traceability effectiveness.

### *Firm ISO Certification*

Established norms and/or codified requirements for a product or service are often established on technical performance or material standards by governmental agencies (e.g., FDA, U.S. Consumer Products Safety Commission) or industry organizations/associations (e.g., International Standards Organization - ISO) (Maruchek et al., 2011). Many industry standards are voluntary and while compliance is expected and supervised, there is no legal action for non-compliance. The most common public standards are those of the International Standards Organization (ISO). Of the ISO standards the most ubiquitous (industry independent and global in nature) are the ISO 9000(1) series of quality standards and the ISO 22000 standard dedicated to management of safety across supply chains (Trienekens & Zuurbier, 2008). Many guidelines require that manufacturers meet ISO standards for compliance (e.g., the FDA's Good Manufacturing Practice - GMP).

However, realized benefits of standards have been mixed with some debate on whether they promote better product/service quality and signal good management or, if they simply promote a more minimalist management style of only doing what is necessary to meet standards and requirements (Corbett et al., 2005; Sroufe & Curkovic, 2008). Ultimately, research finds that ISO does provide some tangible benefits, including enhancing internal and external product/service traceability, developing better transfer of product/service information to suppliers upstream in the supply chain, consolidating audits across the supply chain, and improving supplier's standards (Trienekens & Zuurbier, 2008). Moreover, Henson and Reardon (2005) posit that in addition to ensuring improved traceability and safety, standards also provide a form of differentiation and a competitive advantage when management supports the certification effort. Therefore, we hypothesize:

**Hypothesis One (H1):** Firms that are ISO certified have increased/better \_\_\_\_\_ than those firms, which are not ISO certified.

1. top management support
2. non-conforming processes/procedures
3. lot traceability
4. internal systems management for traceability
5. external systems management for traceability

### *Supplier ISO Certification*

Supply chain managers have an on-going concern about the quality of their suppliers. Supply chain transactions have become increasingly distant as well as global, making it particularly difficult for buyers to observe and manager all of their suppliers' required qualifications. Some firms have begun to look at multinational standards to act as a proxy for unobservable qualifications. Over time, ISO certifications have become part of the criteria for supplier selection (Thaver & Wilcock, 2006). In fact, many firms find that to participate in global markets, ISO certification is a requirement (Terlaak & King, 2006). ISO certifications also can signal to both potential customers and competitors a

favorable operational performance and/or competitiveness as a supplier (Terlaak & King, 2006; Thaver & Wilcock, 2006).

Larger firms with better developed management strategies (e.g., enhanced supplier selection criteria, larger search sets, and more stringent supplier requirements) tend to choose suppliers that are ISO certified (Terlaak & King, 2006). They also argue that meeting ISO certification help ensure compliance with a wide range of quality system requirements (e.g., non-conforming processes/procedures, lot traceability, internal and external systems management). Darnall and Edwards (2006) research affirms that quality-based management systems require an organization-wide commitment to continuous improvement of the firm's products/services. Therefore, we hypothesize:

**Hypothesis Two (H2):** Firms that require their suppliers to be ISO certified have increased/better \_\_\_\_\_ than those firms, which do not require supplier ISO certified.

1. top management support
2. non-conforming processes/procedures
3. lot traceability
4. internal systems management for traceability
5. external systems management for traceability

### *Product/Service Risk*

The lack of traceability can hurt supply chain performance and expose firms to myriad of potentially damaging risks. Firms without traceability initiatives in place face supply chain risks including the inability to coordinate activities along a supply chain that lead to failure and error in supply chain operations (Charlier & Valceschini, 2008), problems to fulfill customer demand because of their inability to predict precise item quantity, and accommodating customer order changes (Lee & Özer, 2007; Chrysoschoidis et al., 2009). Overall, risk management may be viewed as the enclosure of traceability. Benefits of risk mitigation through traceability include:

- Meeting safety and security requirements for recordkeeping and counterfeit detection
- Verifying process certification (e.g., certified organic or certified non-child labor)
- Preventing damage to brand and corporate reputation caused by taking too long to identify and remove contaminated product from the marketplace
- Achieving the required “one up, one back” traceability required by law
- Complying with environmental and import requirements

Certain products and services carry a higher perceived risk, whether due to the nature of the product/service (hazardous materials or classified digital information), or due to possible disruptions/interruptions during transit (Chopra & Sodhi, 2004). Agricultural products, chemicals, sensitive digital or computer-based information (e.g., credit cards), and pharmaceuticals carry more risk than products such as hand towels.

The lack of monitoring of activities and operations in supply chains can also lead to deviation from a preferred standard and hurt product quality, including the inability of firms to prevent raw materials or components in products that may pose a health hazard to consumers (Lyles et al., 2008). Also, traceability lends quantitative, evidentiary support to firms' operations and products. This empowers the development of a sense of security and awareness that they can display an accurate record of their actions that will provide them a safeguard if non-conformance occurs.

Firms often adopt preventative measures through traceability to reduce the potential issues the come with high risk products and services (Clarke, 1997). For example, enhanced process management (e.g., internal and external processes to prevent, detect, respond and recover from an incident) is usually initiated by governments or agreed upon by senior management (e.g., Hazard Analysis and Critical Control Point - HACCP or the Strategic Approach to International Chemicals Management). Information sharing of high-risk products/services and suspicious incidents helps firms respond to and prevent risk-related issues (e.g. lot traceability. In addition, internal and external information sharing ensure congruent communication within the firm and throughout the supply chain. Therefore, we propose the following hypothesis:

**Hypothesis Three (H3):** Firms that have higher risk products/services have increased/better \_\_\_\_\_ than those firms, which have less risky products/services.

1. top management support
2. non-conforming processes/procedures
3. lot traceability
4. internal systems management for traceability
5. external systems management for traceability

### *Supply Chain Complexity*

Supply chain complexity is defined as the level of detail and dynamic nature of an organization's downstream markets (Bozarth et al., 2009). Drivers of complexity include the number of customers in the market, the magnitude of customer relationship tasks, demand management tasks, the average length of the product life cycle, and variability of the demand. These drivers increase as the supply chain with in addition of each supply chain partner, especially downstream. For example, local marketplaces (fewer customers and customer relationships, less variability, etc.) would have low complexity while regional and national marketplaces map to increased complexity and global marketplaces (numerous customers with many customer relationships, and high levels of variability) are considered highly complex.

Firms with more complex supply chains need to have enhanced management oversight, integrate non-conforming processes/procedures more deeply into their product/service traceability requirements, and invest in systems that prevent, detect, respond, and recover from disruptions. Research has established that firms with more complex supply chains are more likely to make investments to manage or even reduce complexity where needed (Perrow, 1984). Furthermore, firms with higher risk supply chains tend to more aggressively manage their supply chain and supply chain partners. Supply chain partners may be held more accountable for their product and service traceability and standardization of operations may be more common so as to reduce complexity and enhance traceability (Speier et al., 2011). Therefore, we hypothesize:

**Hypothesis Four (H4):** Firms with greater supply chain complexity have increased/better \_\_\_\_\_ than those firms, which have less supply chain complexity.

1. top management support
2. non-conforming processes/procedures
3. lot traceability
4. internal systems management for traceability
5. external systems management for traceability



## METHODOLOGY

### Survey Development and Data Collection

An online survey was developed to test the hypotheses. The constructs of traceability effectiveness, top management support, established non-conforming process/procedures, lot traceability, internal system management process for traceability, and external system management process for traceability, were operationalized and measured using items adopted from Speier et al. (2011). The constructs have been shown to satisfy the requirements of reliability and validity in construct measurement (Ketokivi & Schroeder, 2004). The items for all of the constructs in the model utilized a four item, seven-point Likert type scale ranging from 1 (not at all) to 7 (very much).

The target population sample was supply chain managers. Contacts were obtained via two sources: members of the Council for Supply chain Management Professionals (CSCMP) and members of the Institute for Supply Management. A total of 2,044 email addresses were obtained from these organizations. Content and face validity of the survey were demonstrated in two steps. First, supply chain professionals reviewed the questionnaire for readability and ambiguity (Dillman, 2000). Second, a pilot study using 40 random contacts from the sample was conducted. Feedback was used to modify the survey. The survey link was then emailed to contact list and after two reminders, a total of 238 usable questionnaires were obtained, for a response rate of 11.6% percent. Non-response bias was assessed by comparing early vs. late respondents and not deemed a threat to data integrity (Armstrong & Overton 1977). The first and fourth quartiles of respondents were tested for differences in mean construct responses. The results of this analysis indicated that there were no significant differences (p-values between 0.072 and 0.301) in the responses obtained from the two groups.

### *Respondent Demographics*

Respondent demographics provide an interesting snapshot into our sample. For example, 50% were working for publically held firms while 43% indicated they worked for privately held firms with 7% considering themselves working for a local/state/federal governmental entity. Respondents fell into four categories of responsibility: procurement/sourcing (52%), Materials Management/Planning/Production/Quality (15%), Inventory/Warehousing (15%), Service/Sales (10%), and Transportation/Logistics (8%) (Table 1). Procurement spend, a proxy for firm size, indicated that most respondents worked in firms with spend greater than \$40 million (57%), while those working for small firms (\$ spend less than \$0.5 million) accounted for less than 4% of respondents.

An equal percentage of the respondents indicated that their firm was ISO certified or currently pursuing ISO certification (43%) as those who were not ISO certified (43%) while 14% indicated they were unsure regarding certification. It must be noted, of those respondents that stated their organizations were ISO certified or pursuing ISO certification, 70% reported to be ISO 9001:2008 and 30% stated another ISO standard. This is important as no respondents identified their firms to be certified under the newest ISO standard, ISO 9001:2015. This point is elaborated on further in the section entitled limitations and future research area.

The survey data revealed that sixty-three percent of respondents identified their products or services as low risk (less perishable, not easily contaminated, non-hazardous), while 37% identified their products or services as high risk (more perishable, easily contaminated, hazardous). Data analysis also showed that 21% of respondents indicated they have low complexity (low reach or more local/regional supply chain reach), 19% indicated medium complexity (national supply chain reach), while 60% indicated high complexity (global supply chain reach). This categorization matches with the procurement spend demographics reported earlier.

### Data Analysis

The data was assessed through construct validity using tests to measure reliability, normality, convergent validity, and discriminant validity. Reliability was used to test for internal consistency

**Table 1. Respondent demographics**

| <b>Firm Ownership</b>                            |      |
|--------------------------------------------------|------|
| Public                                           | 50%  |
| Private                                          | 43%  |
| Government                                       | 7%   |
| n=203 (some blank responses)                     |      |
| <b>Procurement/Sourcing</b>                      |      |
| Materials Management/Planning/Production/Quality | 15%  |
| Inventory/Warehousing                            | 15%  |
| Service/Sales                                    | 10%  |
| Transportation/Logistics                         | 7%   |
| n=164 (some blank responses)                     |      |
| <b>ISO Certification</b>                         |      |
| Yes                                              | 43%  |
| ISO 9001:2008                                    | 70%  |
| Other ISO Standard                               | 30%  |
| No                                               | 43%  |
| Unsure                                           | 6%   |
| n=227 (some blank responses)                     |      |
| <b>Product/Service Risk</b>                      |      |
| High                                             | 0.37 |
| Low                                              | 0.63 |
| n=227 (some blank responses)                     |      |
| <b>Complexity</b>                                |      |
| High                                             | 60%  |
| Medium                                           | 19%  |
| Low                                              | 21%  |
| n=236 (some blank responses)                     |      |

of the research variables and was assessed using Cronbach's alpha on each construct (Table 2). All estimates exceeded the 0.70 cutoff recommended by Nunnally (1979). The full variance-covariance matrix is included in Table 3 in order to address possible instances of covariation between variables (Shah & Goldstein 2006). To assess the responses for normality, tests for skewness and kurtosis were performed on each of the indicators for each construct. Results indicated normality assumptions were met. Common method bias was evaluated by using a three-item marker variable that was not related in any way to the variables in the model (Podsakoff et al., 2003). When paired with the other constructs in the model, none of the correlations were significant at the 0.05 level. Thus, common method bias was not deemed a problem in the data.

Convergent validity was assessed using standard regression loadings from SPSS. Most items were above the desired item loading of 0.70 and were statistically significant; a few items were retained with loadings between 0.50 - 0.70 in order to preserve theoretical integrity. There were no items loading below 0.50 (Garver & Mentzer, 1999). To assess discriminant validity of the

**Table 2. Survey scale items and reliability analysis; the survey scales used a seven-point Likert type scale ranging from 1 (not at all) to 7 (very much)**

| Construct Items                                                                                                                                                                                 | Mean | Std. Dev. |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----------|
| Top management support (Cronbach's alpha = 0.878)                                                                                                                                               | 5.02 | 1.638     |
| • Views traceability within the supply chain as a strategic priority                                                                                                                            |      |           |
| • Has a dedicated position focusing on traceability within the supply chain                                                                                                                     |      |           |
| • Views traceability within the supply chain as a competitive advantage                                                                                                                         |      |           |
| Nonconforming Processes/Procedures (Cronbach's alpha = 0.936)                                                                                                                                   | 5.49 | 1.34      |
| • Prevent a contamination/quality/non-conformance event in our supply chain                                                                                                                     |      |           |
| • Detect a contamination/quality/non-conformance event in our supply chain                                                                                                                      |      |           |
| • Respond to a contamination/quality/non-conformance event in our supply chain                                                                                                                  |      |           |
| • Recover from a contamination/quality/non-conformance event in our supply chain                                                                                                                |      |           |
| Lot Traceability (Cronbach's alpha = 0.84)                                                                                                                                                      | 5.32 | 1.32      |
| • Origin and destination points/dates of order, receipt/release/service dates, number within order, description of item/service                                                                 |      |           |
| • Routing and transfer points through which a shipment/order/service request moved (e.g., movement from production to distribution to shipping or service initialization to service conclusion) |      |           |
| • External traceability data (i.e., external to your firm or after the product leaves your firm or service is complete)                                                                         |      |           |
| Internal Systems Management (Cronbach's alpha = 0.95)                                                                                                                                           | 5.45 | 1.36      |
| • Provides our supply chain partners with timely data in response to a contamination/quality/non-conformance incident                                                                           |      |           |
| • Provides our supply chain partners with valid data in response to a contamination/quality/non-conformance incident                                                                            |      |           |
| • Communicates with our supply chain partners in a timely manner in response to contamination/quality/non-conformance incidents                                                                 |      |           |
| External Systems Management (Cronbach's alpha = 0.93)                                                                                                                                           | 5.16 | 1.43      |
| • Defines consequences for supply chain partners who fail to comply with supply chain contamination/quality/non-conformance procedures                                                          |      |           |
| • Requires our supply chain partners to provide us the actionable information our firm needs to respond to a contamination/quality/non-conformance incident                                     |      |           |
| • Uses audits to determine if relationships should be maintained with supply chain partners                                                                                                     |      |           |
| • Audits the contamination/quality/non-conformance procedures of our supply chain partners                                                                                                      |      |           |
| • Collaborates with our supply partners to improve their contamination/quality/non-conformance programs                                                                                         |      |           |

variables average variance extracted (AVE) was calculated for all constructs (Fornell & Larcker, 1981; Garver & Mentzer, 1999). These results are displayed in Table 4. Values below 0.50 were evaluated; any non-theoretically important items were dropped. Next, the AVE of each construct was compared to the shared variance between all possible pairs of constructs. Any AVE of a single construct that fell below a shared AVE was then tested via nested models to determine significant differences among the constructs in the theoretical model. All constructs met the conditions for discriminant validity.

**Table 3. Discriminant validity (Fornell-Larcker criterion)**

| Correlation Matrix for Constructs |              |              |              |              |              |
|-----------------------------------|--------------|--------------|--------------|--------------|--------------|
|                                   | 1            | 2            | 3            | 4            | 5            |
| 1                                 | <b>0.825</b> |              |              |              |              |
| 2                                 | 0.565        | <b>0.825</b> |              |              |              |
| 3                                 | 0.562        | 0.560        | <b>0.785</b> |              |              |
| 4                                 | 0.525        | 0.628        | 0.672        | <b>0.788</b> |              |
| 5                                 | 0.396        | 0.519        | 0.516        | 0.691        | <b>0.818</b> |

**Note:** Square root of the AVE on diagonal in bold.

**Table 4. Inter-item covariance**

| Inter-Item Covariance Matrix |              |              |              |              |              |
|------------------------------|--------------|--------------|--------------|--------------|--------------|
|                              | 1            | 2            | 3            | 4            | 5            |
| 1                            | <b>2.672</b> |              |              |              |              |
| 2                            | 1.241        | <b>1.807</b> |              |              |              |
| 3                            | 1.214        | 0.995        | <b>1.747</b> |              |              |
| 4                            | 1.174        | 1.155        | 1.214        | <b>1.868</b> |              |
| 5                            | 0.927        | 0.999        | 0.977        | 1.354        | <b>2.053</b> |

**Note:** Variance on the diagonal in bold

### MANOVA Testing

This research studies multiple dependent variables and therefore used MANOVA to analyze the data. MANOVA was used to examine the effect of the four key factors (ISO certification, ISO certification of suppliers, product risk, and supply chain complexity) on the five constructs of traceability. Significant effects were found among the four key factors (ISO Certification: Wilks'  $L = 0.949$ ,  $F(5,187) = 2.00$ , and a  $p < 0.008$ ); ISO Certification of Suppliers: Wilks'  $L = 0.792$ ,  $F(5,187) = 9.80$ ,  $p < 0.000$ ); Product Risk: Wilks'  $L = 0.917$ ,  $F(5,187) = 3.37$ , and a  $p < 0.006$ ); Supply Chain Complexity: Wilks'  $L = 0.927$ ,  $F(10,374) = 2.97$ ,  $p < 0.013$ ).

MANOVA was then conducted to examine gaps in multiple dependent variables. However, one limitation of this method is its inability to test where the specific differences occurred. Therefore, we followed the initial analysis with the significance test, analysis of variance (ANOVA) to look at individual dependent variables separately. ANOVA is a clear method of capturing the key features of complex and hierarchical data structures or study designs (i.e. the current study has five dependent variables with four independent variables) and presenting key results. The use of ANOVA also avoids the issue with regression of having to specify an over complex model and of carrying multiple pairwise t-tests between the levels of those independent variables with in the model. Given that the MANOVA resulted in significant differences, main effects among the four key factors and the dependent variables (five constructs of traceability) were tested using ANOVA.

### ANOVA Testing

The results of the ANOVA testing revealed that the main effects for ISO Certification, ISO Certification of Suppliers, and Supply Chain Complexity were significant across all five constructs of traceability, thereby fully supporting H1, H2, and H4. The main effect for Product Risk were significant for four

of the five constructs of traceability, thereby partially supporting H3. In the case of ISO Certification, ISO Certification of Suppliers, and Supply Chain Complexity the factors are bi-level or contain Yes/No or High/Low designations and the ANOVA can be interpreted directly from the output. However, in the case of Supply Chain Complexity, the factor has three levels. Thus, we deemed that additional post-hoc tests must be performed to interpret the factor's significance. The next sections will discuss the significance of the ANOVA for each the dependent variables with the post-hoc tests included for the Supply Chain Complexity factor.

### *Implications of ANOVA and Post-Hoc Test Results*

ISO certification of a firm is the independent variable for H1. It is hypothesized that firms that are ISO certified will answer with a higher score on each of the five constructs of traceability than those firms, which are not ISO certified. In support of H1, respondents whose firm was ISO certified tended to give higher rating on the five key constructs of traceability. A more detailed analysis revealed that the ratings for ISO certified firms not only were higher than those of non-ISO certified firms, but the difference was statistically significant at the 0.05 level (Table 5).

ISO certification for a firm's suppliers is the independent variable for H2. It is hypothesized that firms that require their suppliers to be ISO certified will answer with a higher rating on each of the five constructs of traceability than those firms, which do not require ISO certified suppliers. In support of H2, respondents whose firm required ISO certified suppliers tended to give higher rating on the five constructs of traceability. Specifically, the ratings for firms requiring ISO certified suppliers were higher than those of non-ISO certified firms and the difference was statistically significant at the 0.05 level (Table 6).

Hypothesis 3 stated that firms with products/services that are higher in risk will answer with higher rating on each of the five constructs of traceability than those firms, which have lower risk products/services. In support of H3, respondents whose firm identified as having higher risk products/

**Table 5. Mean ratings for ISO vs non-ISO certified firms**

| Construct                            | ISO  |      |
|--------------------------------------|------|------|
|                                      | Yes  | No   |
| C1-Top Management Support            | 5.32 | 4.78 |
| C2-Non-Conforming Process Management | 5.94 | 5.19 |
| C3-Lot Traceability                  | 5.70 | 4.91 |
| C4-Internal Traceability Processes   | 5.91 | 5.31 |
| C5-External Traceability Processes   | 5.68 | 4.75 |

**Table 6. Mean ratings for ISO certified suppliers vs non-ISO certified suppliers**

| Construct                            | ISO Certified Suppliers |      |
|--------------------------------------|-------------------------|------|
|                                      | Yes                     | No   |
| C1-Top Management Support            | 5.29                    | 4.80 |
| C2-Non-Conforming Process Management | 5.77                    | 5.23 |
| C3-Lot Traceability                  | 5.53                    | 4.96 |
| C4-Internal Traceability Processes   | 5.76                    | 5.41 |
| C5-External Traceability Processes   | 5.84                    | 4.55 |

services tended to have higher ratings on four of the five key constructs of traceability. In fact, on those four key constructs the differences between firms with higher risk compared to lower risk products/services were statistically significant at the 0.05 level (Table 7).

Firms with more complex supply chains were hypothesized to have higher rating on the five constructs of traceability than firms with less complex supply chains. Supply chain complexity was representing on a tri-part scale of more complex, medium complex, or low complex. H4 was supported with respondents whose firm identified as having more complex supply chains tended to have higher ratings on the five constructs of traceability. In fact, on those key dimensions the differences between firms with different levels of supply chain complexity were statistically significant at the 0.05 level (Table 8).

Post hoc tests revealed that three constructs (Top Management Support, Non-Conforming Process Management, and Internal Traceability Processes) revealed differences between organizations with more complex supply chains and organizations with less complex supply chains. For these three constructs organizations with more complex supply chains responded with higher ratings and those rating were statistically different from organizations with less complex supply chains.

In the case of Lot Traceability and External Traceability Processes, post hoc tests revealed statistically significant differences between all three levels of supply chain complexity. Specifically, in both constructs the post hoc tests revealed that the pair-wise comparisons of the ratings differences between organizations with more complex supply chains, medium, and less complex supply chains all were statistically significant at the 0.05 level.

**Table 7. Mean ratings for firms with high product/service risk vs low product/service risk**

| Construct                            | Product/Service Risk |      |
|--------------------------------------|----------------------|------|
|                                      | High                 | Low  |
| C1-Top Management Support*           | 5.20                 | 4.94 |
| C2-Non-Conforming Process Management | 5.86                 | 5.33 |
| C3-Lot Traceability                  | 5.66                 | 5.01 |
| C4-Internal Traceability Processes   | 5.94                 | 5.31 |
| C5-External Traceability Processes   | 5.57                 | 4.91 |

\* not statistically significant at the 0.05 level

**Table 8. Mean ratings for firms with more complex supply chains vs less complex supply chains**

| Construct                            | Supply Chain Complexity |                |              |
|--------------------------------------|-------------------------|----------------|--------------|
|                                      | More Complex            | Medium Complex | Less Complex |
| C1-Top Management Support            | 5.25                    | 5.09           | 4.31         |
| C2-Non-Conforming Process Management | 5.73                    | 5.24           | 5.04         |
| C3-Lot Traceability                  | 5.53                    | 5.15           | 4.42         |
| C4-Internal Traceability Processes   | 5.80                    | 5.53           | 4.96         |
| C5- External Traceability Processes  | 5.50                    | 4.83           | 4.45         |

**Note:** using a LSD test for individual means the means in bold differ from one another at the 0.05 level

## CONCLUSION

Organizations today face significant visibility and supply-chain issues due to the length and complexity of global supply chains. These issues can be exacerbated by poorly managed, or non-existent traceability systems within firms and throughout their supply chains. The purpose of this paper was to investigate the path to greater traceability effectiveness vis-à-vis the interaction of a theoretical model of five dimensions of traceability and four key factors that represent potential considerations that may impact a firm's product/service traceability. The behavior of the model was explained using the tenets of contingency theory.

Our primary contribution is that the research finds empirical support for contingency theory in the context of supply chain traceability. The research indicated that the dimensions of top management support, non-conforming processes/procedures, lot traceability, internal systems management, and external systems management all were statistically significant differentiators with regards to firm ISO certification, supplier ISO certification, and supply chain complexity. For the factor defined as product/service risk, four of the five dimensions were statistically significant with top management support as the lone dimension not significant. A summary of each hypothesis and the statistical significance of each dimension is displayed in Table 9. While Table 9 illustrates the overall results of the four hypotheses and the contribution therein, more detail must be provided on each individual hypothesis to add weight to the results of this research.

First and foremost, the results indicate that managers believe firms that are ISO certified and require their suppliers to be ISO certified are more likely to possess the five dimensions of traceability, providing support for H1 and H2. This outcome reinforces the findings from Trienekens and Zuurbier (2008) in that ISO certification, whether it be at the firm level or within a firm's suppliers, does provide tangible benefits. In addition, with H1 and H2 being supported, this research supports Henson and Reardon's (2005) notion that along with ensuring improved traceability and safety, private standards also provide a form of differentiation and a competitive advantage when management supports the certification effort. From a contingency perspective, this finding suggests that managers understand the relationship between macro-level certifications and traceability dimensions and more specifically, how the former influences management decisions to create the latter.

The findings also found support for H3. Excluding the non-statistical result for top management support this research agrees with Clark's (1997) research. Clark's (1997) work establishes that to reduce the potential issues that more risky products/services bring with them, many firms adopt preventative measures such as non-conforming processes/procedure, lot traceability, and internal

Table 9. Summary of Hypotheses Outcome and Statistical Significance

| Traceability Dimensions             | H1 – Firm ISO Certification | H2 – Supplier ISO Certification | H3 – Product /Service Risk | H4 – Supply Chain Complexity |
|-------------------------------------|-----------------------------|---------------------------------|----------------------------|------------------------------|
| Top Management Support              | 0.01                        | 0.02                            | 0.24*                      | 0.00                         |
| Non-Conforming Processes/Procedures | 0.00                        | 0.00                            | 0.00                       | 0.00                         |
| Lot Traceability                    | 0.00                        | 0.00                            | 0.00                       | 0.00                         |
| Internal Systems Management         | 0.00                        | 0.05                            | 0.00                       | 0.00                         |
| External Systems Management         | 0.00                        | 0.00                            | 0.00                       | 0.00                         |

(NOTE: All dimensions are significant for all hypotheses except for Top Management Support in the case of H3 – this hypothesis is denoted with an asterisk)

as well as external systems management. For this study's sample, it appears that top management support is not a differentiator in regard to product/service risk magnitude (low risk vs. high risk).

One possible explanation may lie with variables such as firm size and industry. This study did not control for firm size or industry due to low sample size and the subsequent loss of power within the statistical tests. However, future research should include a control for firm size since larger organizations may have more hierarchical control in their management structure which in turn could lead to differences in top management support. Also, industry type may play a larger role in top management support since some industries like food products (e.g., HACCP) or healthcare products/services (e.g., CDRH) are mandated by certain government/agency policies that may predispose top management to adopt traceability standards therefore innately requiring top management support.

In addition, supply chain traceability originally attracted the attention of public authorities and private companies in high risk industries (i.e., food, food service, pharmaceutical, and health) due to the potential hazards of unsafe products and services. As a result, the U.S. Food and Drug Administration (FDA) and U.S. Department of Agriculture (USDA) regulations have great impact on a firm's supply chain in terms of traceability.

Thus, more and more companies view traceability as part of competitive strategy in the supply chain (Canavari et al., 2010). In turn, top management is willing to support the development of traceability beyond legal requirements and baseline expectations in high risk products/services although establishment of traceability systems is not mandatory. In recent years, public safety is garnering more attention and is being extended to less risky products/services. This is evidenced by the increased number of recalls across a variety of industries such as home appliances, children's products, furniture, mobile devices etc. (Szal, 2016).

Two factors contribute to the increased number of recalls. First, in business today, companies operate in the environment that many of them have interwoven supply chains, resulting in more widespread and complicated recalls. Second, tighter regulation and better detection technologies are applied even in less risky products/services businesses in response to increasing consumer demand on product quality and safety (Szal, 2016). Realizing the increasing scope, complexity and costs of recalls and a poorly executed recall can be devastating to a company's reputation, senior management tends to view supply chain traceability as a strategic priority in order to have a system in place to address procedures and communication along the end-to-end supply chain in case a recall does occur. This non-supported hypothesis reflects the trend that supply chain traceability is in general viewed as a competitive advantage by business executives.

All of the five key dimensions of traceability were supported in H4. This agrees with Perrow's (1984) assertion that firms with more complex supply chains are more likely to make investments to manage or even reduce complexity where needed. The support of H4 also is in agreement with Speier et al. (2011) that supply chain partners may be held more accountable for their product/process traceability and standardization of operations so as to reduce complexity and enhance traceability. The results also suggest that managers believe complexity drives firms to create a contingent response to that complexity in the form of more effective traceability initiatives. Complexity often instills uncertainty in managers, who treat uncertainty with the same strategies as dynamism. Supply chain complexity and product complexity have been recognized as critical contingent factors in supply chain management research (Blome, Schoenherr, & Eckstein, 2014). Prior studies suggest that the complexity of product/process and supply chain complexity complicate decision making, decrease operations performance, and cause disruption (Narasimhan & Talluri, 2009; Hu et al., 2008). Contingent strategies in the form of traceability processes is therefore a reaction to complexity.

Effective traceability can be a valuable strategy to firms, increasing organizations' competitive advantage (Barney, 1991). These competitive skill differences can improve organizations' profits and demonstrate governance value (Ghosh & John, 1999). Furthermore, organizations must not only possess such resources, managers must deploy resources in ways that most effectively fit the given context.



## Limitations and Future Research

Our study has some limitations to be addressed in future research. First, our survey data called for the subjective perceptions of managers in relation to traceability. Although an approach well established in prior studies was used, future research should consider more objective measures including cost, secondary data from customer satisfaction surveys, and other relevant firm metrics. Second, due to the time frame of data collection and the updating of ISO standards (i.e., data collection occurred during late 2016 while the ISO 9001:2015 standard was in its infancy), our data does not include a sampling of organizations using the newer ISO 9001:2015 standard. Future research would need to address traceability through the lens of the ISO 9001:2015 standard and compare to the conclusions obtained using the older ISO 9001:2008 standard. Controls for firm size and industry must also be included in this research area.

Another area of future research is the impact of contracts and contracting on traceability. There has been limited work on insurance and contracting with suppliers especially for forward traceability. Bulut and Lawrence (2008) both found a negative association with the use of contracting and the adoption of traceability practices. However, their studies only applied to the food supply chain. Ramesch and Jarke (2001) studied the information technology supply chain and found that subcontracting brings difficulties with tracing which entity provided which parts and whose parts affects which other parts of the subcontracted project. Future research is needed to address other industries and industry wide standards for how contracting effects traceability in the supply chain.

## REFERENCES

- Armstrong, J. S., & Overton, T. S. (1977). Estimating nonresponse bias in mail surveys. *JMR, Journal of Marketing Research*, 14(3), 396–402. doi:10.2307/3150783
- Barney, J. (1991). Firm resources and sustained competitive advantage. *Journal of Management*, 17(1), 99–120. doi:10.1177/014920639101700108
- Barratt, M., & Choi, T. (2007). Mandated RFID and institutional responses: Cases of decentralized business units. *Production and Operations Management*, 16(5), 569–585. doi:10.1111/j.1937-5956.2007.tb00281.x
- Bechini, A., Cimino, M. G., Marcelloni, F., & Tomasi, A. (2008). Patterns and technologies for enabling supply chain traceability through collaborative e-business. *Information and Software Technology*, 50(4), 342–359. doi:10.1016/j.infsof.2007.02.017
- Bertolini, M., Bevilacqua, M., & Massini, R. (2006). FMECA approach to product traceability in the food industry. *Food Control*, 17(2), 137–145. doi:10.1016/j.foodcont.2004.09.013
- Blackhurst, J., Craighead, C. W., Elkins, D., & Handfield, R. B. (2005). An empirically derived agenda of critical research issues for managing supply-chain disruptions. *International Journal of Production Research*, 43(19), 4067–4081. doi:10.1080/00207540500151549
- Blome, C., Schoenherr, T., & Eckstein, D. (2014). The impact of knowledge transfer and complexity on supply chain flexibility: A knowledge-based view. *International Journal of Production Economics*, 147, 307–316. doi:10.1016/j.ijpe.2013.02.028
- Bosona, T., & Gebresenbet, G. (2013). Food traceability as an integral part of logistics management and agricultural supply chain. *Food Control*, 33(1), 32–48. doi:10.1016/j.foodcont.2013.02.004
- Bozarth, C. C., Warsing, D. P., Flynn, B. B., & Flynn, E. J. (2009). The impact of supply chain complexity on manufacturing plant performance. *Journal of Operations Management*, 27(1), 78–93. doi:10.1016/j.jom.2008.07.003
- Bulut, H., & Lawrence, J. D. (2008). Meat slaughter and processing plants' traceability levels: evidence from Iowa. *Economics Working Papers*, 168.
- Canavari, M., Centonze, R., Hingley, M., & Spadoni, R. (2010). Traceability as part of competitive strategy in the fruit supply chain. *British Food Journal*, 112(2), 171–186. doi:10.1108/00070701011018851
- Charlier, C., & Valceschini, E. (2006). Traceability, trust and coordination in a food chain. In *Trust and Risk in Business Networks*. Bonn: ILB.
- Chen, C., Zhang, J., & Delaurentis, T. (2014). Quality control in food supply chain management: An analytical model and case study of the adulterated milk incident in China. *International Journal of Production Economics*, 152, 188–199. doi:10.1016/j.ijpe.2013.12.016
- Chopra, S., & Sodhi, M. S. (2004). Managing risk to avoid supply-chain breakdown. *MIT Sloan Management Review*, 46(1), 53.
- Chrysoschoidis, G., Karagiannaki, A., Pramataris, K., & Kehagia, O. (2009). A cost-benefit evaluation framework of an electronic-based traceability system. *British Food Journal*, 111(6), 565–582. doi:10.1108/00070700910966023
- Clarke, R. V. G. (Ed.). (1997). *Situational crime prevention*. Monsey, NY: Criminal Justice Press.
- Corbett, C. J., Montes-Sancho, M. J., & Kirsch, D. A. (2005). The financial impact of ISO 9000 certification in the United States: An empirical analysis. *Management Science*, 51(7), 1046–1059. doi:10.1287/mnsc.1040.0358
- Dai, H., Tseng, M. M., & Zipkin, P. H. (2015). Design of traceability systems for product recall. *International Journal of Production Research*, 53(2), 511–531. doi:10.1080/00207543.2014.955922
- Darnall, N., & Edwards, D. (2006). Predicting the cost of environmental management system adoption: The role of capabilities, resources and ownership structure. *Strategic Management Journal*, 27(4), 301–320. doi:10.1002/smj.518
- Dillman, D. A. (2000). *Mail and internet surveys: The tailored design method* (Vol. 2). New York: Wiley.

- Donaldson, L. (2001). The contingency theory of organizations. *Sage (Atlanta, Ga.)*.
- Ellis, S. C., Henry, R. M., & Shockley, J. (2010). Buyer perceptions of supply disruption risk: A behavioral view and empirical assessment. *Journal of Operations Management*, 28(1), 34–46. doi:10.1016/j.jom.2009.07.002
- ECR Europe. (2004). *ECR-using traceability in the supply chain to meet consumer safety expectations*.
- Fearne, A. (1998). The evolution of partnerships in the meat supply chain: Insights from the British beef industry. *Supply Chain Management*, 3(4), 214–231. doi:10.1108/13598549810244296
- Fornell, C., & Larcker, D. F. (1981). Evaluating structural equation models with unobservable variables and measurement error. *JMR, Journal of Marketing Research*, 18(1), 39–50. doi:10.2307/3151312
- Garver, M. S., & Mentzer, J. T. (1999). Logistics research methods: Employing structural equation modeling to test for construct validity. *Journal of Business Logistics*, 20(1), 33.
- Germain, R., Davis-Sramek, B., Lonial, S. C., & Raju, P. S. (2011). The impact of relational supplier exchange on financial performance: A study of the hospital sector. *Journal of Business Logistics*, 32(3), 240–253. doi:10.1111/j.2158-1592.2011.01020.x
- Ghosh, M., & John, G. (1999). Governance value analysis and marketing strategy. *Journal of Marketing*, 63, 131–145. doi:10.2307/1252107
- Grötsch, V. M., Blome, C., & Schleper, M. C. (2013). Antecedents of proactive supply chain risk management—a contingency theory perspective. *International Journal of Production Research*, 51(10), 2842–2867. doi:10.1080/00207543.2012.746796
- Gunasekaran, A., & Ngai, E. W. (2004). Virtual supply-chain management. *Production Planning and Control*, 15(6), 584–595. doi:10.1080/09537280412331283955
- Handfield, R. B., Cousins, P. D., Lawson, B., & Petersen, K. J. (2015). How can supply management really improve performance? A knowledge-based model of alignment capabilities. *The Journal of Supply Chain Management*, 51(3), 3–17. doi:10.1111/jscm.12066
- Henson, S., & Reardon, T. (2005). Private agri-food standards: Implications for food policy and the agri-food system. *Food Policy*, 30(3), 241–253. doi:10.1016/j.foodpol.2005.05.002
- Hu, J., Zhang, X., Moga, L. M., & Neculita, M. (2013). Modeling and implementation of the vegetable supply chain traceability system. *Food Control*, 30(1), 341–353. doi:10.1016/j.foodcont.2012.06.037
- Hu, S. J., Zhu, X., Wang, H., & Koren, Y. (2008). Product variety and manufacturing complexity in assembly systems and supply chains. *CIRP Annals - Manufacturing Technology*, 57(1), 45–48.
- International Organization for Standardization. (2007). New ISO Standard to Facilitate Traceability in Food Supply Chains ISO 22005:2007.
- Jakkhupan, W., Arch-int, S., & Li, Y. (2015). An RFID-based traceability system: A case study of rice supply chain. *Telecommunication Systems*, 58(3), 243–258. doi:10.1007/s11235-014-9866-7
- Jansen-Vullers, M. H., van Dorp, C. A., & Beulens, A. J. (2003). Managing traceability information in manufacture. *International Journal of Information Management*, 23(5), 395–413. doi:10.1016/S0268-4012(03)00066-5
- Kärkkäinen, M., Ala-Risku, T., & Främling, K. (2003). The product centric approach: A solution to supply network information management problems? *Computers in Industry*, 52(2), 147–159. doi:10.1016/S0166-3615(03)00086-1
- Kelepouris, T., Pramataris, K., & Doukidis, G. (2007). RFID-enabled traceability in the food supply chain. *Industrial Management & Data Systems*, 107(2), 183–200. doi:10.1108/02635570710723804
- Kleindorfer, P. R., & Saad, G. H. (2005). Managing disruption risks in supply chains. *Production and Operations Management*, 14(1), 53–68. doi:10.1111/j.1937-5956.2005.tb00009.x
- Kumar, S., Heustis, D., & Graham, J. (2015). The future of traceability within the U.S. food industry supply chain: A business case. *International Journal of Productivity and Performance Management*, 64(1), 129–146. doi:10.1108/IJPPM-03-2014-0046

- Lee, H., & Özer, Ö. (2007). Unlocking the value of RFID. *Production and Operations Management*, 16(1), 40–64. doi:10.1111/j.1937-5956.2007.tb00165.x
- Luthans, F., & Stewart, T. I. (1977). A general contingency theory of management. *Academy of Management Review*, 2(2), 181–195. doi:10.5465/amr.1977.4409038
- Lyles, M. A., Flynn, B. B., & Frohlich, M. T. (2008). All supply chains don't flow through: Understanding supply chain issues in product recalls. *Management and Organization Review*, 4(2), 167–182. doi:10.1111/j.1740-8784.2008.00106.x
- MacCarthy, B. L., Blome, C., Olhager, J., Singh, S., & Zhao, X. (2016). Supply chain evolution—theory, concepts and science. *International Journal of Operations & Production Management*, 36(12), 1696–1718. doi:10.1108/IJOPM-02-2016-0080
- Malles, D. A., Sawhney, R., & De Anda, E. M. (2015). Implementation of serialization and traceability in a pharmaceutical packaging company. In *IIE Annual Conference Proceedings*, Norcross.
- Maruchek, A., Greis, N., Mena, C., & Cai, L. (2011). Product safety and security in the global supply chain: Issues, challenges and research opportunities. *Journal of Operations Management*, 29(7), 707–720. doi:10.1016/j.jom.2011.06.007
- Mattevi, M., & Jones, J. A. (2016). Food supply chain: Are UK SMEs aware of concept, drivers, benefits and barriers, and frameworks of traceability? *British Food Journal*, 118(5), 1107–1128. doi:10.1108/BFJ-07-2015-0261
- Merminod, N., & Paché, G. (2011). Supply management and corporate social responsibility: The challenge of global chain traceability. *Journal on Chain and Network Science*, 11(3), 213–222. doi:10.3920/JCNS2011.x201
- Mode, P. F. (1988). *Effects Analysis in Design (Design FMEA) and for Manufacturing and Assembly Processes (Process FMEA) Instruction Manual*. Ford Motor Company.
- Narasimhan, R., & Talluri, S. (2009). Perspectives on risk management in supply chains. *Journal of Operations Management*, 27(2), 114–118. doi:10.1016/j.jom.2009.02.001
- Nunnally, J. C. (1979). *Nunally Psychometric theory*. New York: McGraw-Hill.
- Peri, C. (2002). Rintracciabilità della filiera dei prodotti agroalimentari: significato, metodi e strumenti. *Rintracciabilità di filiera per una provincia trasparente Indagine sulla filiera ortofrutticola*, Atti del convegno, Cuneo, 6.
- Perrow, C. (1984). *Normal Accidents: living With High-Risk Technologies*. Princeton, New Jersey: Princeton University Press.
- Podsakoff, P. M., MacKenzie, S. B., Lee, J. Y., & Podsakoff, N. P. (2003). Common method biases in behavioral research: A critical review of the literature and recommended remedies. *The Journal of Applied Psychology*, 88(5), 879–903. doi:10.1037/0021-9010.88.5.879 PMID:14516251
- Ramesh, B., & Jarke, M. (2001). Toward reference models for requirements traceability. *IEEE Transactions on Software Engineering*, 27(1), 58–93. doi:10.1109/32.895989
- Regattieri, A., Gamberi, M., & Manzini, R. (2007). Traceability of food products: General framework and experimental evidence. *Journal of Food Engineering*, 81(2), 347–356. doi:10.1016/j.jfoodeng.2006.10.032
- Rinehart, L. M., Myers, M. B., & Eckert, J. A. (2004). Supplier relationships: The impact on security. *Supply Chain Management Review*, 8(6), 52–59.
- Ruekert, R. W., Walker, O. C. Jr, & Roering, K. J. (1985). The organization of marketing activities: A contingency theory of structure and performance. *Journal of Marketing*, 49(1), 13–25. doi:10.2307/1251172
- SAE. (2015). *AS5678A: Passive RFID Tags Intended for Aircraft Use in G-18 Radio Frequency Identification (RFID)*.
- Sarpong, S. (2014). Traceability and supply chain complexity: Confronting the issues and concerns. *European Business Review*, 26(3), 271–284. doi:10.1108/EBR-09-2013-0113

- Senneset, G., Forås, E., & Fremme, K. M. (2007). Challenges regarding implementation of electronic chain traceability. *British Food Journal*, 109(10), 805–818. doi:10.1108/00070700710821340
- Shafiq, A., Klassen, R. D., Johnson, P. F., & Awaysheh, A. (2014). Socially responsible practices: An exploratory study on scale development using stakeholder theory. *Decision Sciences*, 45(4), 683–716. doi:10.1111/deci.12085
- Shah, R., & Goldstein, S. M. (2006). Use of structural equation modeling in operations management research: Looking back and forward. *Journal of Operations Management*, 24(2), 148–169. doi:10.1016/j.jom.2005.05.001
- Silvestro, R., & Lustrato, P. (2014). Integrating financial and physical supply chains: The role of banks in enabling supply chain integration. *International Journal of Operations & Production Management*, 34(3), 298–324. doi:10.1108/IJOPM-04-2012-0131
- Skilton, P. F., & Robinson, J. L. (2009). Traceability and normal accident theory: How does supply network complexity influence the traceability of adverse events? *The Journal of Supply Chain Management*, 45(3), 40–53. doi:10.1111/j.1745-493X.2009.03170.x
- Speier, C., Whipple, J. M., Closs, D. J., & Voss, M. D. (2011). Global supply chain design considerations: Mitigating product safety and security risks. *Journal of Operations Management*, 29(7), 721–736. doi:10.1016/j.jom.2011.06.003
- Sroufe, R., & Curkovic, S. (2008). An examination of ISO 9000: 2000 and supply chain quality assurance. *Journal of Operations Management*, 26(4), 503–520. doi:10.1016/j.jom.2007.06.006
- Steven, A. B. (2015). Supply Chain Structure, Product Recalls, and Firm Performance: Empirically Investigating Recall Drivers and Recall Financial Performance Relationships. *Decision Sciences*, 46(2), 477–483. doi:10.1111/deci.12135
- Szal, A. (2016). Report indicates rising number of recalls by U.S. authorities. *IMPOMag*. Retrieved from <https://www.impomag.com/data-focus/2016/08/report-indicates-rising-number-recalls-us-authorities?cmpid=horizontalent>
- Terlaak, A., & King, A. A. (2006). The effect of certification with the ISO 9000 Quality Management Standard: A signaling approach. *Journal of Economic Behavior & Organization*, 60(4), 579–602. doi:10.1016/j.jebo.2004.09.012
- Thaver, I., & Wilcock, A. (2006). Identification of overseas vendor selection criteria used by Canadian apparel buyers: Is ISO 9000 relevant? *Journal of Fashion Marketing and Management: An International Journal*, 10(1), 56–70. doi:10.1108/13612020610651123
- Thompson, J. D. (1967). *Organizations in action: Social science bases of administrative theory*. New York, NY: Transaction Publishers.
- Trienekens, J., & Zuurbier, P. (2008). Quality and safety standards in the food industry, developments and challenges. *International Journal of Production Economics*, 113(1), 107–122. doi:10.1016/j.ijpe.2007.02.050
- U.S. House of Representatives. (2007). Reducing Fraudulent and Imitation Drugs Act of 2007, H.R. 2716, 110th Cong. Retrieved from <http://www.govtrack.us/congress/bill.xpd?bill=h110-2716>
- Department of Defense, US Military. (1980). US Military Standard, MIL-STD-1629A (1980). Procedures for performing a failure mode, effect and criticality analysis.
- Department of Defense, US Military. (1983). US Military Standard, MIL-STD-1629A (1983). Procedures for performing a failure mode, effect and criticality analysis.
- Veronneau, S., & Roy, J. (2009). RFID benefits, costs, and possibilities: The economical analysis of RFID deployment in a cruise corporation global service supply chain. *International Journal of Production Economics*, 122(2), 692–702. doi:10.1016/j.ijpe.2009.06.038
- Viaene, J., & Verbeke, W. (1998). Traceability as a key instrument towards supply chain and quality management in the Belgian poultry meat chain. *Supply Chain Management*, 3(3), 139–141. doi:10.1108/13598549810230868
- Voss, M. D., Whipple, J. M., & Closs, D. J. (2009). The role of strategic security: Internal and external security measures with security performance implications. *Transportation Journal*, 48(2), 5–23.

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